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# Steering Control for Autonomously Balancing Bicycle at Low Speed

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*Abstract*— **This paper studies how to use steering control to keep the bicycle balanced at low speed (< 2m/s). By changing the trail length of the bicycle and designing a corresponding low-speed balance control algorithm, the bicycle can retain the favorable characteristics of simple structure and high efficiency. Firstly, we construct the bicycle dynamic model, and analyze the influence of different trail lengths on the dynamic characteristics of the bicycle. Then, we design the steering control algorithm. The results of both the simulation and experiments on prototypes show that the proposed algorithm can meet the requirements of balance control when the bicycle is running at low speed. After changing the trail length, the minimum speed of the bicycle prototype is reduced from 1.3m/s to 0.58m/s with good stability.**

## I. INTRODUCTION

In the past decade, autonomous technology has made great breakthroughs, and autonomous vehicles are gradually moving from experimental stage to commercialization. Currently, autonomous vehicles are represented by four-wheeled vehicles such as cars. Compared with a car, a single-track vehicle like a bicycle is another common type of transportation in life. It has the advantages of low price, flexibility, lightness and energy saving, which is the preferred mean of transportation for people on short-distance trips. Therefore, it is equally significant to realize autonomous driving of bicycles.

However, bicycles cannot be self-balanced when no one is manipulating. Therefore, in order to realize autonomous driving of bicycles, balance control must first be achieved. Although the research on the balance control of bicycles and motorcycles has been going on for nearly 60 years[1], until now, the technology is not perfect and still in the research and experimental stage. Affected by this difficulty, autonomous bicycle is still a brand new field.

There are many related studies on the balance control of bicycles, and different researchers propose different methods. The essence of these methods is to generate a balance torque that can resist gravitational torque by some means, so as to avoid the falling of the bicycle. According to the way of generating the balance torque, the balance control methods can be divided into three types: inverted pendulum control[2], moment gyro control[3], and steering control. The first two methods can balance the bicycle at stationary and have good stability, but they also need to install additional devices on the bicycles to generate the balance torque, which increase the energy consumption and structural complexity.

The steering control method changes the centrifugal torque (when the bicycle has speed) and the additional torque of the front fork by rotating the handlebar to balance the bicycle. Unlike the first two methods, steering control utilizes the dynamic characteristics of the bicycle itself, with a simple structure and high efficiency. Many researchers have realized the constant-speed steering control of bicycles[4], [5], [6], [7]. On the basis of previous research, He[8] used the speed segmentation method and neural network method to achieve the varying-speed balance control of the bicycle. In the above studies, the minimum speed of the bicycle is about 5∼7km/h.

When the bicycle is stationary, it is very difficult to achieve balance control by this method because there is no centrifugal force, so there are few related studies at present. In 2011, Yi et al[9] first tried to use steering control to achieve automatic balance when the motorcycle is stationary. The prototype could maintain balance at stationary but has low stability.

In 2017, Honda changed the motorcycle's trail length by moving the front wheel forward, thus changing the dynamic characteristics of the motorcycle, realizing the automatic balance of the motorcycle at low speed and at stationary with good stability[10]. However, Honda did not public its technical details.

Therefore, this paper plans to change the trail length and achieve the automatic balance of the bicycle at low speed using steering control method so the bicycle can retain the favorable characteristics of simple structure and high efficiency. Compared with motorcycles, bicycles have higher COG(center of gravity) and finer tires, making it more difficult to achieve low-speed balance control.

The rest of the paper is organized as follows: Section II introduces the dynamic model of the bicycle, and quantitatively analyzes the influence of different trail lengths on the dynamic characteristics; Section III designs the low-speed steering control algorithm of the bicycle and the simulation results show its effectiveness. Section IV performs the experimental results of the prototypes. Conclusions for our work and discussion for possible future work are finally provided in Section V.

## II. MODELING & ANALYSIS OF BICYCLE DYNAMICS

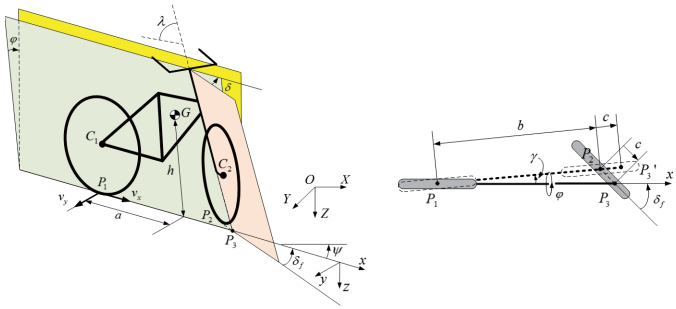
The three-dimensional structure of a normal bicycle is shown in Fig.1(a), and the top view is shown in Fig.1(b). In order to make the formula more concise, we make $c_\varphi := \cos\varphi$, $s_\varphi := \sin\varphi$.

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(a) (b)

Fig. 1. (a) 3D sutructure of bicycle and (b) Top view of bicycle

As is shown in Fig.1, $P_1$ and $P_2$ are the ground contact points of the rear wheel and the front wheel, and $P_3$ is the intersection between the handlebar rotation axis and the ground. The ground coordinate system is established as $O-XYZ$ and the bicycle coordinate system is $P_1-xyz$. In the bicycle coordinate system, the $x$ axis is the extension of $P_1P_3$; $a$ is the horizontal distance from the bicycle's COG to the center of the rear wheel $P_1$, and $b$ is the distance between the front wheel and rear wheel. $c$ is the trail length, which is the distance between $P_2P_3$, $h$ is the height of the COG, $\lambda$ is the front fork angle, and $v_x$ and $v_y$ are respectively The forward speed and lateral speed of the rear wheel, $\varphi$ is the roll angle of the bicycle, $\delta$ is the steering angle of the handlebar, the effective steering angle $\delta_f$ is the projection of $\delta$ on the ground, which has the following relationship:

$$\tan\delta_f c_\varphi = \tan\delta s_\lambda \tag{1}$$

In this paper, the Lagrangian equation is used to model the dynamics of the bicycle. For a normal bicycle, the expression of its Lagrangian function is as follows:

$$L = \frac{1}{2}m\boldsymbol{v}_G{\boldsymbol{v}_G}^T + \frac{1}{2}J_s\dot{\delta}^2 - mg\left(hc_\varphi - \Delta_{h_G}\right) \tag{2}$$

Where $m$ is the mass of the bicycle, and $\Delta_{h_G}$ is the decreased height of COG when rotating the handlebar, $\boldsymbol{v}_G$ is the speed of the COG.

As can be seen from Fig.1(b), due to the presence of the trail, when the handlebar is rotating, $P_1$ and $P_2$ are not changed since the tire slip is not considered, the position of $P_3$ changes and the bicycle moves from the dotted line position to the solid line position. Therefore, the bicycle's body will rotate an angle $\gamma$, which satisfies the following relationship:

$$\frac{b}{s_{\delta_f}} = \frac{c}{s_\gamma} \tag{3}$$

Using the approximation relationship we can get:

$$\gamma \approx s_\gamma \approx \frac{c\cdot\delta_f}{b} \tag{4}$$

In addition, the rotation of the bicycle's body also causes an extra small change in the COG, denoted as $\Delta_{h_G}$. According to [11], $\Delta_{h_G}$ is calculated as follows:

$$\Delta_{h_G} = \gamma a s_\varphi \approx \frac{acs_\lambda s_\varphi}{bc_\varphi}\delta \tag{5}$$

According to [7], the speed of the COG $\boldsymbol{v}_G$ is:

$$\boldsymbol{v}_G = \boldsymbol{v}_{P_1} + \boldsymbol{v}_{P_1G} = \left[v_x - h\dot{\psi}s_\varphi, v_y - a\dot{\psi} - h\dot{\varphi}c_\varphi, h\dot{\varphi}s_\varphi\right] \tag{6}$$

Substituting formula (5) and (6) into formula (2), the bicycle's Lagrangian function $L$ can be expressed as:

$$\begin{aligned} L =& \frac{1}{2}m\left[\left(v_x - hs_\varphi\dot{\psi}\right)^2 + \left(v_y - a\dot{\psi} - h\dot{\varphi}c_\varphi\right)^2 + \left(h\dot{\varphi}s_\varphi\right)^2\right] \\ &+ \frac{1}{2}J_s\dot{\delta}^2 - mg\left(hc_\varphi - \frac{acs_\lambda s_\varphi}{bc_\varphi}\delta\right) \end{aligned} \tag{7}$$

We define generalized speed: $\dot{\boldsymbol{r}} = \left[\dot{\varphi}, v_x, \dot{\delta}\right]^T$, $\dot{\boldsymbol{s}} = [\dot{\psi}, v_y]^T$. According to [7], when the rear wheel has no side slip, it satisfies the following nonholonomic constraint:

$$\dot{\boldsymbol{s}} + \boldsymbol{A}(\boldsymbol{r},\boldsymbol{s})\dot{\boldsymbol{r}} = 0 \tag{8}$$

Where $\boldsymbol{A} = \begin{bmatrix} 0 & -\frac{bc_\varphi}{\delta s_\lambda} & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Substituting the constraint (8) into formula (7) and eliminating the variables in $\dot{\boldsymbol{s}}$, we can get the constrained Lagrangian function $L_c$:

$$\begin{aligned} L_c =& \frac{1}{2}J_s\dot{\delta}^2 - mg\left(hc_\varphi - \frac{acs_\lambda s_\varphi}{bc_\varphi}\delta\right) \\ &+ \frac{m}{2}\left[\left(v_x - \frac{hs_\lambda s_\varphi}{bc_\varphi}\delta v_x\right)^2 + \left(\frac{as_\lambda}{bc_\varphi}\delta v_x + h\dot{\varphi}c_\varphi\right)^2 + \left(h\dot{\varphi}s_\varphi\right)^2\right] \end{aligned} \tag{9}$$

For autonomous bicycles, there are two external inputs: the forward driving force to the rear wheel $F_v$ and the external torque to the handlebar $T_\delta$, so the corresponding generalized force $\boldsymbol{\tau}$ can be expressed as:

$$\boldsymbol{\tau} = [0, F_v, T_\delta]^T \tag{10}$$

Since the rear wheel's motion satisfies the constraint (8), Lagrangian equation cannot be directly used, according to [12], in this case the constrained Lagrange dynamics equation should be used, as can be seen in formula (11):

$$\begin{gathered} \frac{d}{dt}\frac{\partial L_c}{\partial \dot{\boldsymbol{r}}^i} - \frac{\partial L_c}{\partial \boldsymbol{r}^i} + \boldsymbol{A}_i^k\frac{\partial L_c}{\partial \boldsymbol{s}^k} = -\frac{\partial L}{\partial \dot{\boldsymbol{s}}^l}\boldsymbol{B}_{ij}^l\dot{\boldsymbol{r}}^j + \boldsymbol{\tau}^i \\ i,j = 1,2,3 \quad l,m = 1,2 \end{gathered} \tag{11}$$

Where $\boldsymbol{B}_{ij}^l = \left(\frac{\partial \boldsymbol{A}_i^l}{\partial \boldsymbol{r}^j} - \frac{\partial \boldsymbol{A}_j^l}{\partial \boldsymbol{r}^i} + \boldsymbol{A}_i^k\frac{\partial \boldsymbol{A}_j^l}{\partial \boldsymbol{s}^k} - \boldsymbol{A}_j^k\frac{\partial \boldsymbol{A}_i^l}{\partial \boldsymbol{s}^k}\right)$.

Substituting the formula (7), (8), (9) and (10) into the constrained Lagrange equation (11), the following three dynamic equations can be derived:

$$\begin{aligned} &mh^2\ddot{\varphi} + \frac{mahs_\lambda}{b}\delta\dot{v}_x - mghs_\varphi + \frac{mahs_\lambda}{b}v_x\dot{\delta} - \frac{macgs_\lambda}{bc_\varphi^2}\delta \\ &+ \frac{mhv_x^2s_\lambda\delta}{b}\left(1 - \frac{h\delta s_\lambda s_\varphi}{bc_\varphi}\right) + \frac{mahv_xs_\lambda s_\varphi}{bc_\varphi}\delta\dot{\varphi} = 0 \end{aligned} \tag{12}$$



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$$m\dot{v}_x\left(1-\frac{2h\delta s_\varphi s_\lambda}{bc_\varphi}+\frac{h^2\delta^2 s_\lambda^2 s_\varphi^2}{b^2c_\varphi^2}+\frac{a^2\delta^2 s_\lambda^2}{b^2c_\varphi^2}\right)+\frac{mahs_\lambda\delta}{b}\ddot{\varphi}$$
$$-\frac{2mhv_x\dot{\varphi}\delta s_\lambda}{bc_\varphi^2}\left(1-\frac{h\delta s_\lambda s_\varphi}{bc_\varphi}\right)+\frac{ma^2s_\lambda^2v_x\delta}{b^2c_\varphi^2}\left(\frac{s_\varphi}{c_\varphi}\delta\dot{\varphi}+\dot{\delta}\right)$$
$$+\frac{mhv_x\delta\dot{\varphi}s_\lambda s_\varphi^2}{bc_\varphi^2}\left(1-\frac{h\delta s_\lambda s_\varphi}{bc_\varphi}\right)+\frac{mh^2v_x\delta\dot{\delta}s_\lambda^2s_\varphi^2}{b^2c_\varphi^2}$$
$$-\frac{mhv_xs_\lambda s_\varphi}{bc_\varphi}\dot{\delta}-\frac{mah\dot{\varphi}^2\delta s_\lambda s_\varphi}{bc_\varphi}=F_v \tag{13}$$

$$J_s\ddot{\delta}-\frac{macgs_\lambda s_\varphi}{bc_\varphi}=T_\delta \tag{14}$$

Where the equation (12) represents the balance dynamics of the bicycle, the equation (13) represents the forward dynamics and the equation (14) represents the steering dynamics.

For the convenience of analysis, this paper considers the dynamic characteristics of the bicycle when the speed is constant. Therefore, the forward dynamics (13) can be ignored. By linearizing the remaining two equations, the simplified bicycle dynamics can be obtained:

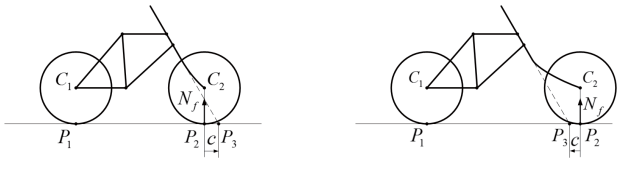
$$\begin{cases} mgh\varphi-mh^2\ddot{\varphi}+\frac{mahs_\lambda}{b}v_x\dot{\delta}+\left(\frac{mhv_x^2s_\lambda}{b}-\frac{macgs_\lambda}{b}\right)\delta=0 \\ J_s\ddot{\delta}-\frac{macgs_\lambda}{b}\varphi=T_\delta \end{cases} \tag{15}$$

As can be seen from the dynamics (15), there are three torques that mainly affect the balance characteristics of the bicycle:

- The gravitational torque $T_G=mgh\varphi$, which is related to roll angle $\varphi$;
- The centrifugal torque $T_{v_x}=\frac{mhv_x^2s_\lambda}{b}\delta$, which is related to steering angle $\delta$ and foward speed $v_x$;
- The additional torque of the front fork $T_c=-\frac{macgs_\lambda}{b}\delta$, which is related to steering angle $\delta$ and trail length $c$.

Since the trail length is small, $T_c$ is much smaller than $T_{v_x}$ when the bicycle is running at high speed. When the speed is very low, $T_{v_x}$ is small, so the bicycle needs to perform a larger steering to resist the gravitational torque, and in this case, $T_c$ is particularly important.

When $P_3$ is before $P_2$, the trail length $c$ is positive, and when $P_3$ is after $P_2$, the trail length $c$ is negative. The structures of the two types of bicycles are shown in Fig.2.




(a) (b)

Fig. 2. Structures of bicycles when (a) $c$ is positive and (b) $c$ is negative

Next, we will analyze the influence of trail length on the dynamic characteristics of the bicycle at low speed.

In the dynamics (15), we let the external input $T_\delta=0$, and set the speed $v_x=0.6$m/s. At the initial moment, the bicycle has a roll angle to the left ($\varphi=1^\circ$). Through simulation, we can obtain the handlebar's minimum initial steering angular velocity that makes the bicycle back to upright position with different trail lengths. The results are shown in Fig.3.

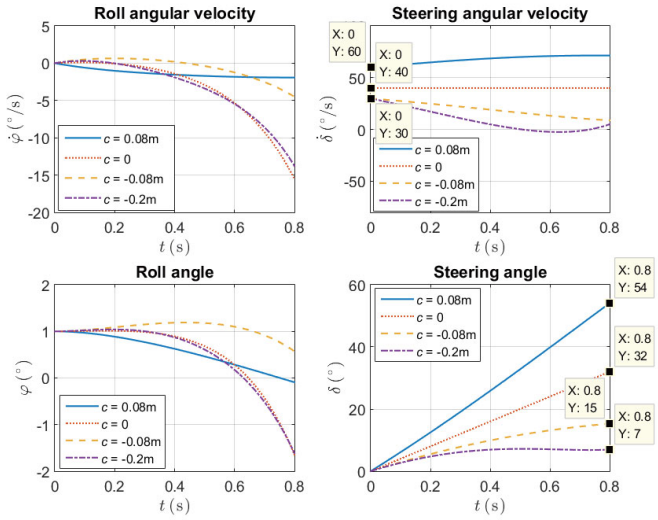



Fig. 3. bicycle dynamic characteristics simulation results

From Fig.3 we can see that when the trail length is 0.08m, the handlebar needs to rotate at least $60^\circ/$s to make the bicycle back to upright position. At 0.8s, it rotates $54^\circ$. When the trail length is 0, the minimum angular velocity is reduced to $40^\circ/$s, and the handlebar rotates $32^\circ$ at 0.8s. When the trail lengths are $-0.08$m and $-0.2$m, the handlebar's minimum angular velocity and the steering angle are further reduced, and only $15^\circ$ and $7^\circ$ are respectively rotated at 0.8s.

In summary, when the trail length is positive, the directions of the front fork additional torque $T_c$ and the centrifugal torque $T_{v_x}$ are always opposite so their effects are mutually weakened. When the speed is very low, the handlebar needs to perform a rapid and large steering action to resist gravitational torque, the stability of the bicycle is very poor. When the trail length is negative, the directions of $T_c$ and $T_{v_x}$ are the same, and their effects are mutually enhanced. Thus, the handlebar requires only a small steering action to make the bicycle back to upright position, and the stability of the bicycle at low speed is improved. Therefore, changing the bicycle's trail length makes it possible to realize automatic balance at low speed using steering control.

## III. LOW-SPEED STEERING CONTROLLER DESIGN & SIMULATION

In the bicycle prototype, the steering motor uses the position mode control, the input of the system is the desired steering angle instead of the torque of the handlebar. The position control of the motor can make the actual steering angle follow the desired value well so the steering dynamics (14) can be ignored.

Therefore, when the bicycle's moving speed is constant, the dynamic model can be further simplified to a second-order transfer function between the steering angle $\delta$ and the



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roll angle $\varphi$:

$$G(s)=\frac{\varphi(s)}{\delta(s)}=\frac{-ahs_\lambda v_x s+\left(acg-hv_x^2\right)s_\lambda}{bh^2s^2-ghb} \tag{16}$$

Since the system has a negative real pole, it is open-loop unstable. Besides, since the transfer function has no zero-pole cancellation, the open-loop system is completely controllable, and the system can be closed-loop stabilized by adding a controller. In the closed-loop system, the input of the system is the desired roll angle $\varphi_d$, the output of the controller is the desired steering angle $\delta_d$, and the output of the closed-loop system is the actual roll angle $\varphi$.

Since the bicycle can be simplified to a linear SISO system, the Proportional-integral-derivative(PID) plus feed-forward control algorithm is used to realize the balance control.

In the control algorithm, the PD controller allows the bicycle to respond quickly and suppress the overshoot, and the feedforward controller can make the bicycle quickly reach the desired stable position. According to the transfer function (16), when the bicycle is in a stable state, the steering angle is proportional to the roll angle. Therefore, the value of the theoretical feedforward coefficient can be directly determined by the transfer function, that is:

$$K_F=K_0=\frac{ghb}{hv_x^2s_\lambda-acgs_\lambda} \tag{17}$$

However, due to the difference between the dynamic model and the actual system, the calculated theoretical feedforward coefficient may have a small deviation from the actual system's value. Therefore, the integral controller is added to eliminate the remaining static error. Through feedforward control supplemented with integral control, the response of the system can be both fast and accurate.

Therefore, the steering control law of the bicycle is as follows:

$$\delta_d=K_P\cdot(\varphi_d-\varphi)-K_D\cdot\dot{\varphi}+K_I\cdot\int(\varphi_d-\varphi)+K_F\cdot\varphi_d \tag{18}$$

The structure of the control system is shown in Fig.4:

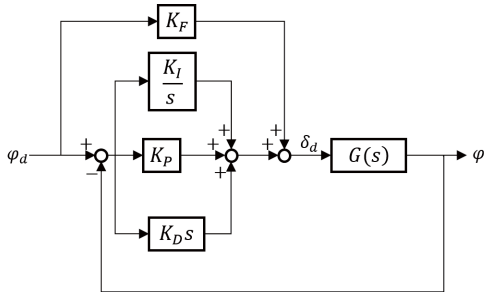
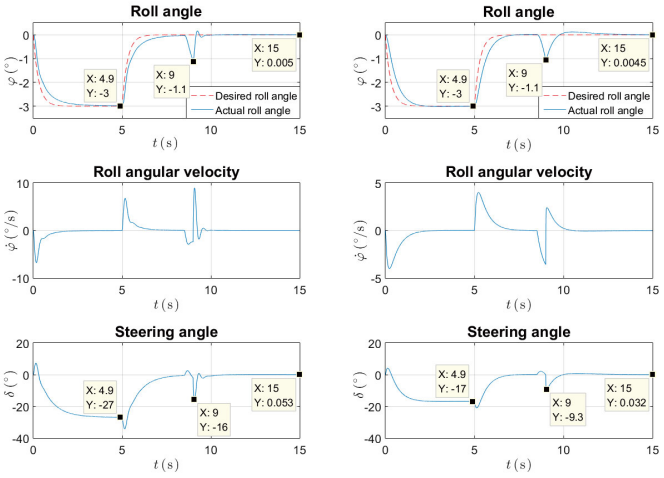



Fig. 4. Steering control system structure of bicycle

According to the Routh criterion, the closed-loop system can be stabilized by selecting reasonable control parameters.

The closed-loop system is simulated using the Matlab Simulink platform. We select two different trail lengths (0.08m, −0.09m) and adjust control parameters to compare the effects of trail length on the steering control.

First, let $v_x=1.3\text{m/s}$. Initially, the bicycle's roll angle $\varphi$ and steering angle $\delta$ are both 0, changing the desired roll angle and applying disturbance to the bicycle, and the simulation results are shown in Fig. 5.




(a) (b)

Fig. 5. Simulation results when (a) $c=0.08\text{m}$, $v_x=1.3\text{m/s}$ and (b) $c=-0.09\text{m}$, $v_x=1.3\text{m/s}$

The bicycle can achieve balance control with both trail lengths, but when the trail length is negative, since the theoretical feedforward coefficient is reduced from 9 to 5.6, the steering angle changes from 27° to 17° with the same roll angle, so the flexibility of steering is reduced. However, when the bicycle is subjected to the same disturbance, the steering angle is reduced from 16° to 9.3°, and the bicycle's stability is improved.

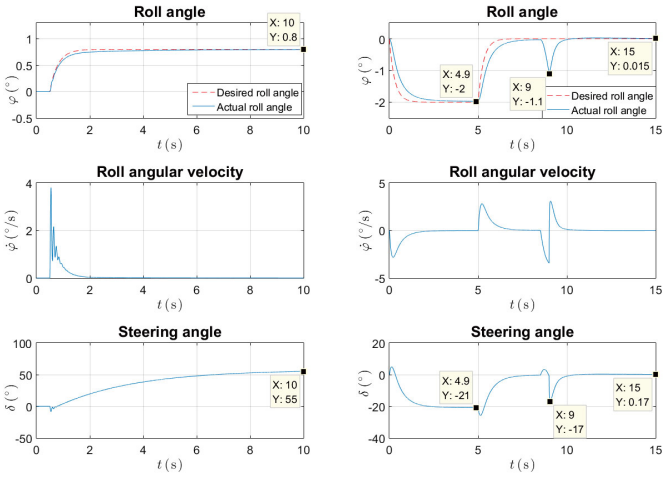
Next, let $v_x=0.8\text{m/s}$. The simulation results are shown in Fig.6. When the trail length is 0.08m, the centrifugal torque is small, and the front fork additional torque further weakens the effect of the centrifugal torque, so the theoretical feedforward coefficient is too large, which is 72.13. When the desired roll angle is only 0.8°, the handlebar needs to rotate 55° in order to maintain balance, and the system's response is very slow, the bicycle cannot achieve balance control. When the trail length is −0.09m, the front fork additional torque enhances the centrifugal torque, and the theoretical feedforward coefficient is reduced to 10.55. The bicycle can maintain balance and has good stability.

From the formula (17), (18) and simulation results, we can find that the key parameter deciding the balance control characteristic of the bicycle is the theoretical feedforward coefficient $K_F$. The larger the coefficient is, the balance control is more difficult. When $K_F$ is greater than a certain level, the balance control cannot be realized. On the contrary, the smaller the coefficient is, the system's stability gets better, but the flexibility of bicycle's steering will gradually reduce. Fig.7 shows the relationship between $K_F$ and $v_x$ under different trail lengths.

By estimating we can find that if the bicycle can achieve balance control, its value of $K_F$ needs to be less than 20,



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(a) (b)

Fig. 6. Simulation results when (a) $c = 0.08$m, $v_x = 0.8$m/s and (b) $c = -0.09$m, $v_x = 0.8$m/s

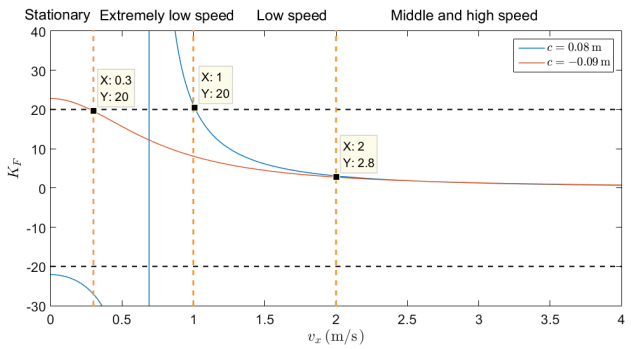



Fig. 7. The relationship between $K_F$ and $v_x$ under different trail length

based on which, for the bicycle prototype in this paper, its forward speed $v_x$ can be divided into four segments:

- When $v_x$ is greater than 2m/s, the value of $K_F$ is very close under different trail lengths. Therefore, the change of the trail length doesn't affect the performance of balance control.
- When $v_x$ is between 1∼2m/s, it is defined as Low speed segment. The bicycle can achieve balance control under different trail lengths. When the trail length is negative, the stability of the bicycle is improved, but the flexibility of steering is weakened.
- When $v_x$ is between 0.3∼1m/s, it is defined as Extremely low speed segment. In this segment, the bicycle with positive trail length cannot maintain balance due to the excessive value of $K_F$, and the bicycle with negative trail length can achieve balance control.
- When $v_x$ is less than 0.3m/s, it is defined as Stationary segment. Since the value of $K_F$ is too large, the balance control cannot be achieved for bicycles with both trail lengths.

Therefore, the low-speed situation of the bicycle discussed in this paper includes Low speed segment and Extremely low speed segment(0.3m/s∼2m/s). The algorithm designed in this section is able to achieve low speed balance control of the bicycle.

## IV. EXPERIMENTAL RESULTS ON PROTOTYPES

In this section, we will verify the steering and anti-interference characteristics of the bicycle prototypes under different speeds and trail lengths through experiments.

In order to carry out the comparative experiment, two bicycle prototypes were built. One still uses the bicycle's own front fork structure, the trail length is 0.08m, which is named Prototype 1. The other prototype can manually adjust the trail length by mechanical modification. The range of the trail length is $-0.01$∼$-0.19$m, which is named Prototype 2. Apart from this, the control system and the mechanical system of the two bicycle prototypes are the same. Fig.8 shows the two prototypes.

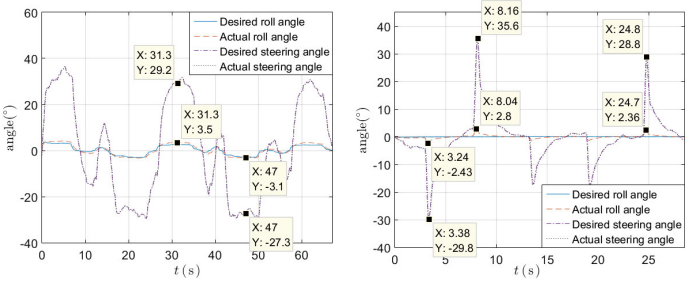



(a) (b) (c)

Fig. 8. (a) Bicycle Prototype 1, (b) Bicycle Prototype 2 and (c) different trail lengths of Prototype 2

In the prototypes, $\varphi$ and $\dot{\varphi}$ are measured by AHRS (Attitude and heading reference system), and $\delta$ is measured by the absolute encoder. $\varphi_d$ is given by the remote controller.

First, let the speed of prototype be 1.3m/s, when the trail length is 0.08m, the experimental results are shown in Fig.9.




(a) (b)

Fig. 9. (a)steering experiment result and (b) anti-interference experiment result when $c = 0.08$m, $v_x = 1.3$m/s

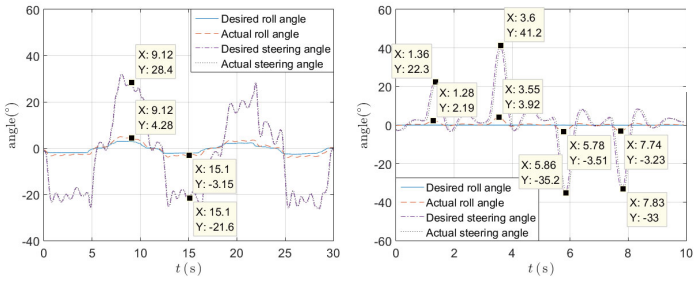
From Fig.9(a) we can see that the actual roll angle can follow the desired roll angle well, and the effect of balance control is good. When the bicycle is in a stable steering state, the ratio of the steering angle to the roll angle is about 8.5, that is, the theoretical feedforward coefficien is about 8.5 at this speed and the trail length. It can be seen from the Fig.9(b) that the bicycle is pushed 5 times in the experiment, and the ratio of the maximum steering angle to the maximum roll angle is about 12.5 when the bicycle is pushed, that is, when the bicycle's roll angle is disturbed by 1°, the handlebar needs to rotate 12.5° to resist the disturbance. After the disturbance disappears, the bicycle prototype can gradually



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return to the upright position, which also indicates that the self-balancing bicycle has anti-interference ability based on steering control.

When the trail length is $-0.09$m, the experimental results are as follow:




(a) (b)

Fig. 10. (a)steering experiment result and (b) anti-interference experiment result when $c = -0.09$m, $v_x = 1.3$m/s

As shown in Fig.10(a), the theoretical feedforward coefficient of the bicycle prototype is about 6.6, which is smaller than the former, and the steering flexibility is reduced. As shown in Fig.10(b), the bicycle is pushed 4 times in the experiment. When the bicycle is pushed, the ratio of the maximum steering angle to the maximum roll angle is about 10.2, which is also smaller than the former, this means that when the bicycle is disturbed by the same amplitude, the handlebar can rotate a smaller angle to resist the disturbance, so the bicycle's stability is improved.

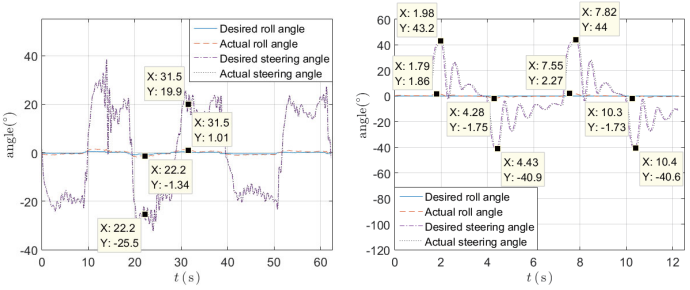
Next, let the speed be 0.58m/s, and when the trail length is 0.08m, the experiment shows that the balance control of the bicycle prototype cannot be realized, which is the same as the simulation result in Section III.

When the trail length is $-0.09$m, the experimental results are shown in Fig.11. At this speed, the range of roll angle is much smaller, but the actual roll angle can still follow the desired value well through steering control, and the bicycle prototype can maintain balance at this speed. When the bicycle is in a stable steering state, the theoretical feedforward coefficient is about 19.5. As shown in figure 11(b), when the bicycle is pushed, the ratio of the maximum steering angle to the maximum roll angle is about 22. The ratio increases much compared to $v_x = 1.3$m/s, which also shows that the disturbance resistance performance of the bicycle will be reduced as the speed decreases. Therefore, the lower the speed, the greater the difficulty of bicycle balance control.

According to the experiments above, we can see that for the normal structure of the bicycle($c = 0.08$m), the minimum speed of balance control is 1.3m/s. By changing the trail length from positive to negative, the bicycle prototype can maintain balance when the minimum speed is 0.58m/s using steering control, and the effectiveness of the control algorithm is verified.

## V. CONCLUSIONS AND DISCUSSION

In this paper, we establish the dynamic model of bicycle, and quantitatively analyze the influence of different speeds




(a) (b)

Fig. 11. $c = -0.09$m, $v_x = 0.58$m/s (a)steering experiment result and (b) anti-interference experiment result

and trail lengths on the dynamic characteristics of bicycle. It is verified that changing the trail length from positive to negative can effectively improve the stability of bicycle when running at low speed. Subsequently, two bicycle prototypes are built and the steering control algorithm is designed. After changing the trail length, the bicycle prototype can achieve balance control with the minimum speed of 0.58m/s and has good stability.

In the future, we plan to use the higher-precision Attitude and Heading Reference System and adopt other more advanced control algorithms to make the bicycle maintain balance at stationary only using steering control, so that the bicycle is closer to realizing autonomous driving.

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